

1 PyNNLF: Python for Network Net Load Forecasting

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5 Summary

6 As solar photovoltaic (PV) system installations increase, network operators must forecast net
7 electricity load—the difference between electricity consumption and PV generation. The term
8 net load forecasting was first introduced in an academic paper in 2016 ([Kaur et al., 2016](#)).
9 Since then, over 60 journal articles and conference papers have been published on the topic by
10 2025 ([Tziolis et al., 2025](#)). Research focus has been mainly on proposing new and complex
11 models, claiming superior performance. Typical statements include:

Statement	Reference
... and it is concluded that the proposed method has higher prediction accuracy and better prediction effect ...	(Cao et al., 2023)
Comparative tests utilizing real-world data verify the superiority of the proposed method over other state-of-the-art algorithms	(Hu et al., 2024)
The performance of the BDLSTM model dominates when compared with the best of the state-of-the-art methods ...	(Sun et al., 2020)

12 However, around 75% of these studies did not use simple benchmark models such as the
13 naïve model, which forecasts the next value as equal to the last observation. Additionally,
14 63% of these studies did not use public datasets, and 99% did not share their code publicly.
15 This indicates a strong focus on improving forecasting accuracy, but limited attention to
16 standardizing evaluation process, model reliability and reproducibility. PyNNLF (Python for
17 Network Net Load Forecasting) is an open-source tool designed to address these gaps by
18 enabling reliable and reproducible evaluation of net load forecasting models. It includes:

19 A library of commonly used net load datasets (e.g., Ausgrid Solar Home Data ([Ausgrid, 2014](#))),
20 and a collection of 18 forecasting models, ranging from simple benchmarks (e.g., naïve
21 model) to statistical models (e.g., linear regression) and machine learning models (e.g.,
22 artificial neural networks).

23 The PyNNLF software is available as an open-source repository on GitHub [here](#) ([Samhan et al., 2025b](#)). Comprehensive documentation is provided [here](#) ([Samhan et al., 2025a](#)).

25 Users can specify the forecasting problem (dataset and forecast horizon) and model configuration
26 (model name and hyperparameters). PyNNLF then outputs evaluation results including
27 performance metrics, metadata, visualizations, and supplemental outputs.

28 Researchers and network operators can use PyNNLF to benchmark their models against
29 others using standardized datasets. They can also contribute new models or datasets to the
30 PyNNLF library, enabling broader comparison and collaboration While libraries like statsmodels,
31 PyTorch, or Darts allow some model comparison with shared data and metrics, they are not
32 designed specifically for net load forecasting. They lack curated datasets and models for this

33 purpose and do not offer a clear way to record experiments. PyNNLF addresses these gaps by
34 providing a focused framework with integrated datasets and structured experiment tracking.

35 In parallel with developing PyNNLF, we are also preparing other research papers: a literature
36 review of net load forecasting studies, and comparative analyses of various models on multiple
37 net load datasets, forecast horizons, spatial aggregations for the load, and minimum demand
38 forecasting using PyNNLF.

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